



**Combined CPS Energy Reactive Power
Planning Project & CPS Energy Helotes
345/138-kV Switching Station and
Autotransformer Addition at Eastside
Switching Station Project – ERCOT
Independent Review Study Status
Update**

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RPG Meeting
October 28, 2025

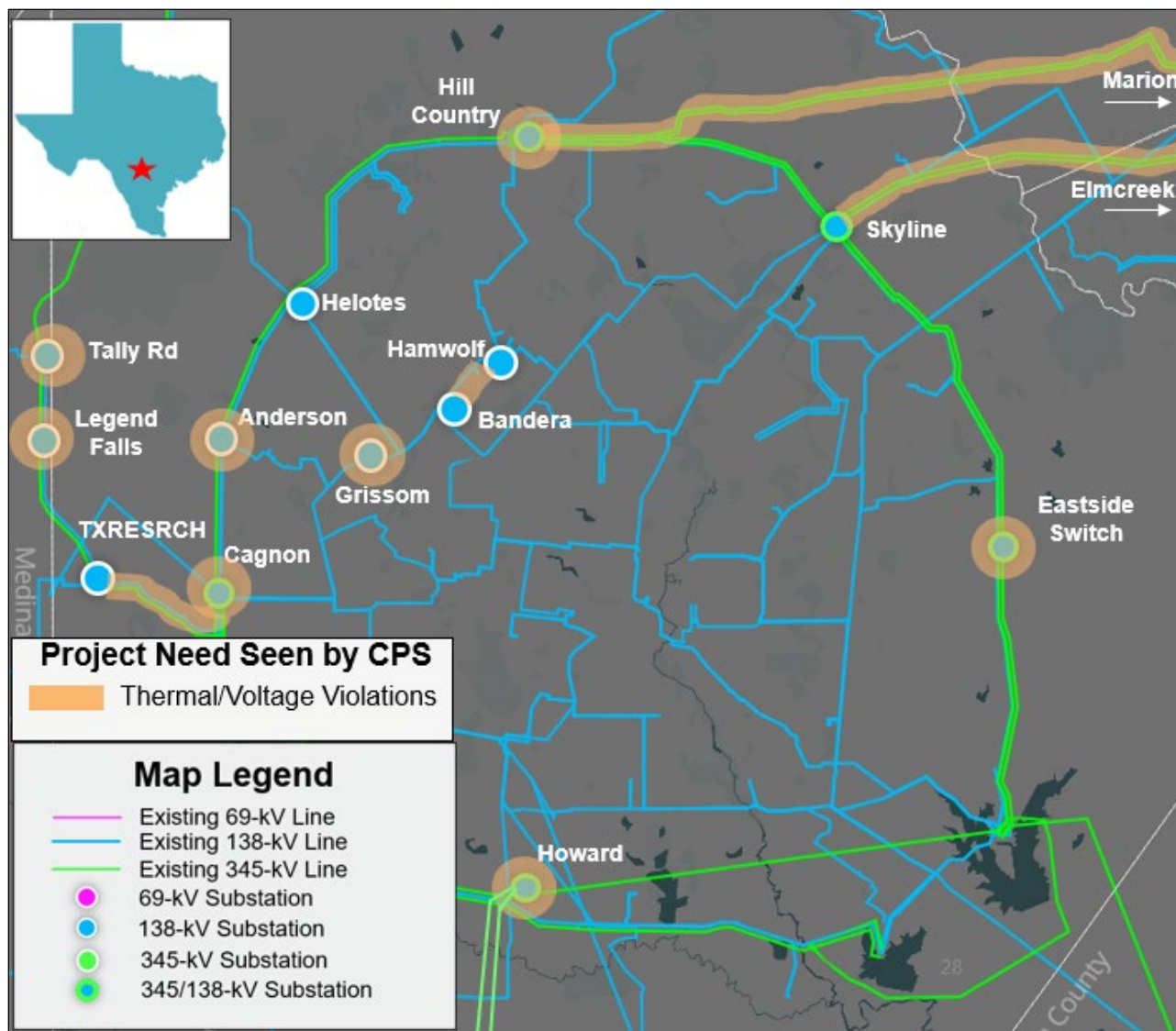
Introduction

- CPS Energy Reactive Power Planning Project (25RPG013) for Regional Planning Group (RPG) review in May 2025
 - This is a Tier 1 project with an estimated cost of \$116.50 million and will not require a Certificate of Convenience and Necessity (CCN)
 - Estimated in-service date (ISD) is December 2029
 - This project is needed to address post-contingency voltage violations in the Bexar County
- CPS Energy Helotes 345/138-kV Switching Station and Autotransformer Addition at Eastside Switching Station Project (25RPG017) for Regional Planning Group (RPG) review in May 2025
 - This is a Tier 1 project with an estimated cost of \$110.0 million and will not require a Certificate of Convenience and Necessity (CCN)
 - Estimated ISD is Summer 2028 and Summer 2029
 - This project is needed to address post-contingency thermal violations in the Bexar County

Introduction cont'd

- CPS presented a project overview and ERCOT provided a project scope at the July 2025 RPG Meeting
 - <https://www.ercot.com/calendar/07292025-RPG-Meeting>
- ERCOT provided a project update at the August RPG Meeting
 - <https://www.ercot.com/calendar/08262025-RPG-Meeting>
- ERCOT is currently conducting a single ERCOT Independent Review (EIR) by combining these two projects (25RPG013 and 25RPG017)

Study Area Map with Project Needs Seen by CPS



Update - Study Assumptions – Generation

- New generation that met the Planning Guide Section 6.9(1) condition with Commercial Operation Date (COD) before June 2030 in and outside the study area at the time of the study will be added to the case if not already modeled in the RTP case based on June 2025 Generator Interconnection Status (GIS) report published in MIS in May 2025
 - Link: <https://www.ercot.com/gridinfo/resource>
 - See Appendix C for the list of generation projects added to the case
- All generation will be dispatched consistent with the 2024 RTP methodology
- All recent retired/indefinitely mothballed units will be reviewed and opened (turned off), if not already reflected in the 2024 RTP Final cases

Update - Study Assumptions – Load & Reserve

- Loads in study area
 - New loads in the study area were added to the study area
 - Approximately 5140 MW of load was added to the study area
- Reserve
 - The reserve will be maintained consistent with the 2024 RTP

Update - Contingencies & Criteria

- Contingencies
 - NERC TPL-001-5.1 and ERCOT Planning Criteria
 - (<http://www.ercot.com/mktrules/guides/planning/current>)
 - P0 (System Intact)
 - P1, P2-1, P7 (N-1 conditions)
 - P2-2, P2-3, P4, and P5 (345-kV only)
 - P3 (G-1+N-1: G-1 of Leon Creek U1 and Guadalupe Energy Center CTG 1)
 - P6-2 (X-1+N-1: X-1 of Cagnon X1, Howard Road X1, and Hill Country X1 345/138-kV transformers)
- Criteria
 - Monitor all 69-kV and above busses, transmission lines, and transformers in the study region (excluding generator step-up transformers)
 - Thermal
 - Use Rate A for normal conditions
 - Use Rate B for emergency conditions
 - Voltage
 - Voltages exceeding their pre-contingency and post-contingency limits
 - Voltage deviations exceeding 8% on non-radial load buses

Update - Preliminary Results of Reliability Assessment – Updated Base Case

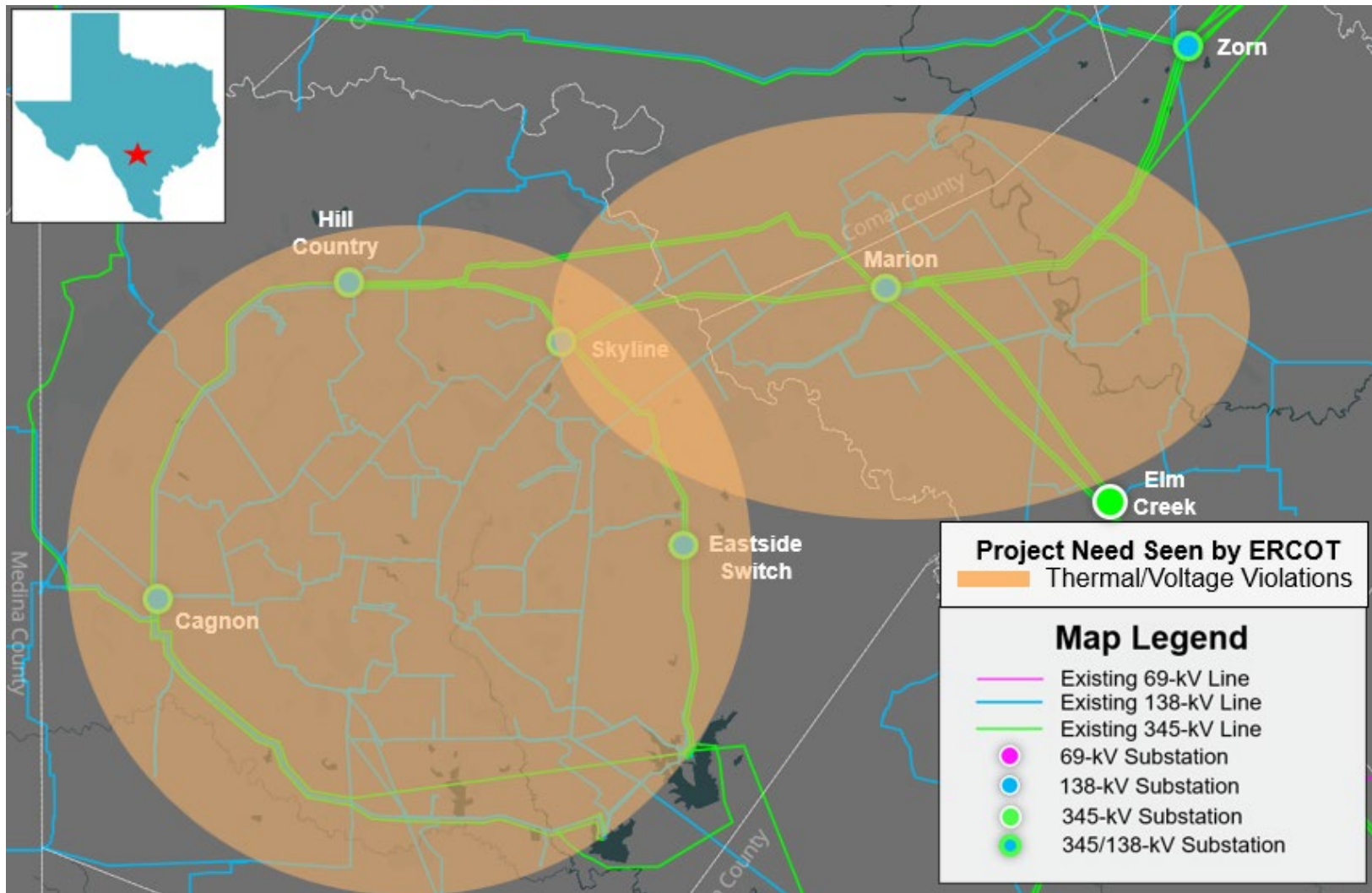
Contingency Category	Unsolved Power Flow	Voltage Violations	Thermal Overloads
P1***	10	1	5
P2, P4, P5	5	None	None
P3 (G-1+N-1)*	45	None	3
P6.2 (X-1+N-1)**	None	None	5
P7***	31	150+	12
Total	91	150+	25

*G-1 Generators tested: Leon Creek U1 and Guadalupe Energy Center CTG 1

**X-1 Transformers tested: Cagnon X1, Howard Road X1, and Hill Country X1 345/138-kV transformers

***Violations seen in the basecase under P1 and P7 events were also seen under G-1+N-1 and X-1+N-1 events

Update - Study Area Map with Project Needs Seen by ERCOT



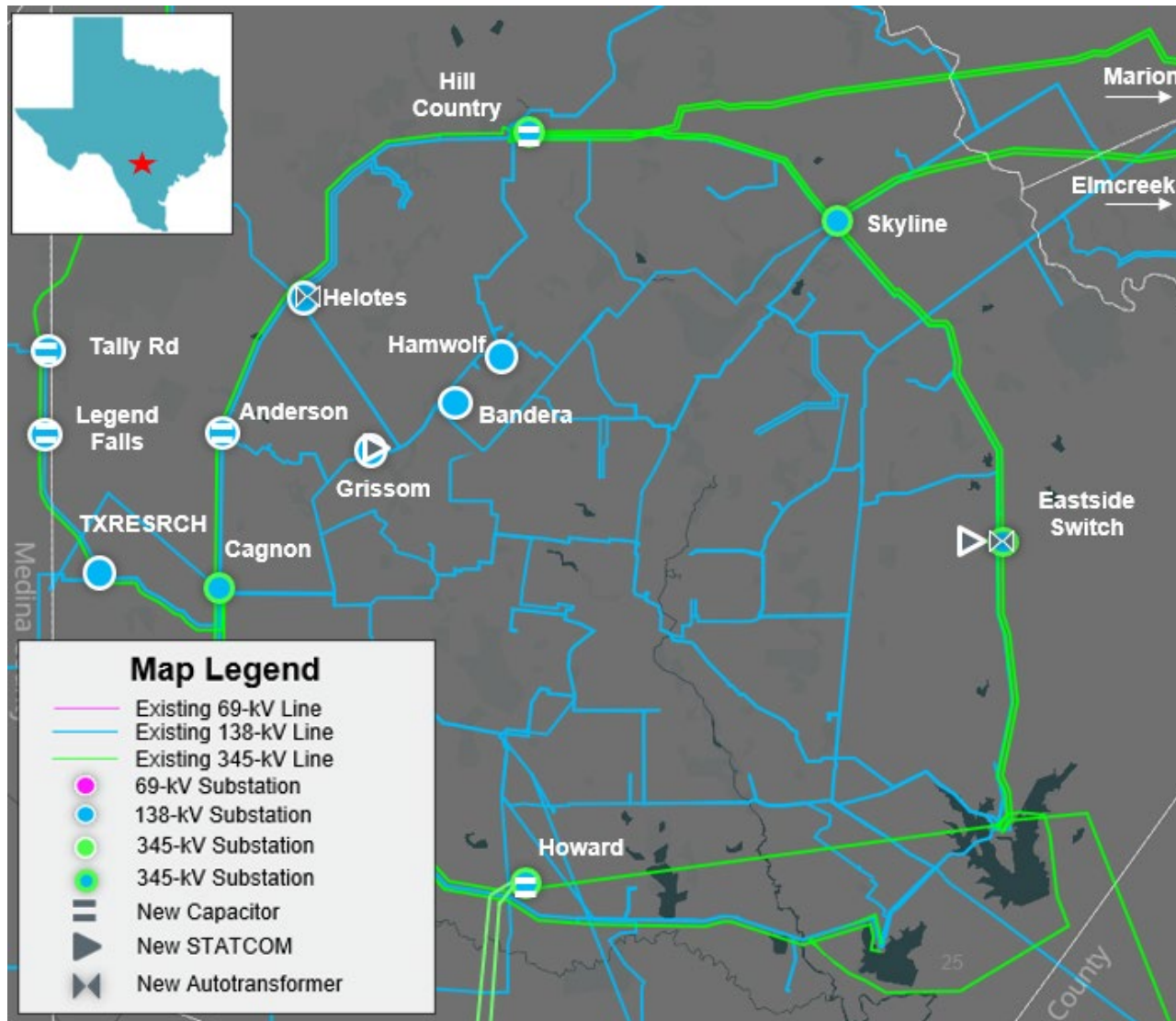
Option 1 – Combined CPS Project

- Install the following sized capacitors at the following substations:
 - 50 MVAR, Hill Country
 - 50 MVAR, Anderson
 - 25 MVAR, Legend Falls
 - 50 MVAR, Howard
 - 25 MVAR, Talley Rd
- Install the following STATCOMs at the following substations:
 - 300 MVAR, Grissom
 - 300 MVAR, Eastside Switch

Option 1 – Combined CPS Project cont'd

- Construct a new Helotes 345-kV substation;
- Convert the existing Helotes 138-kV substation to a new 345/138-kV switching substation;
- Install two new 345/138-kV autotransformers at the new Helotes 345/138-kV switching substation, normal and emergency ratings of at least 600 MVA;
- Loop in the existing Hill Country to Cagnon 345-kV transmission line into the new Helotes 345-kV substation; and
- Install a new 345/138-kV autotransformer at the existing Eastside 345/138-kV substation, normal and emergency ratings of at least 600 MVA

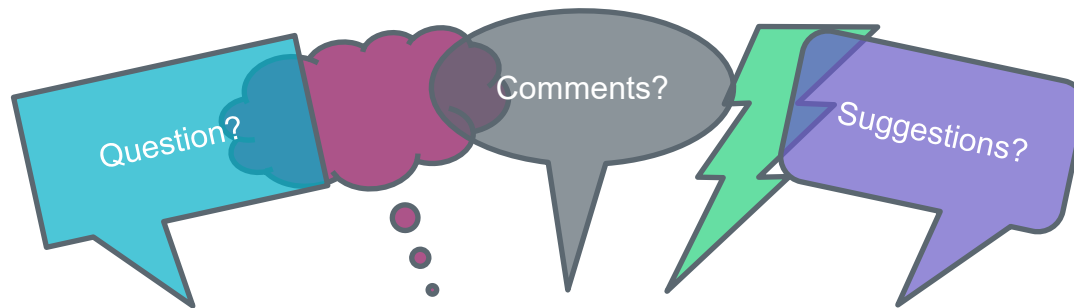
Option 1 – Combined CPS Project



Next Steps and Tentative Timeline

- ERCOT will continue with project evaluation and perform the following
 - Conduct Planned Maintenance Outage Evaluation
 - Conduct Long-Term Load-Serving Capability Assessment
 - Request TSPs to conduct Cost Estimate and Feasibility Assessment
- Additional analyses to be performed on the preferred option
 - Congestion Analysis to ensure that the identified transmission upgrades do not result in new congestion within the study area
 - Generation Addition and Load Scaling Sensitivity Analyses
 - Planning Guide Section 3.1.3(4)
 - Subsynchronous Resonance (SSR) Assessment
 - Nodal Protocol Section 3.22.1.3(2)
- Tentative Timelines
 - Status updates at the future RPG meetings
 - Final recommendation – Q4 2025

Thank you!



Stakeholder comments also welcomed through:

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Appendix A – Transmission Projects Added

TPIT/RPG No	Project Name	Tier	Project ISD	TSP	County(s)
89912	GVEC_Olmos to Wilson 138 kV TL, T297	Tier 4	12/01/26	GVEC	Guadelupe

Appendix B – Transmission Projects Removed

RTP Project ID	Project Name	TSP(s)	County(s)
2024-SC19	Hillje (44200) to Zorn (7042) 345-kV Line Upgrades	CNP, AEN, LCRA TSC	Wharton, Fayette, Bastrop, Caldwell, Guadalupe
2024-SC26	Shaula (5380) to Elm Creek (5133) to Cachena (5068) 345-kV Line Upgrades	CPS	Wilson, Guadalupe, Dewitt

Appendix C – Generation Added

GINR	Project Name	Fuel	Project COD	Capacity (MW)	County
20INR0162	Diamondback Solar	SOL	12/31/2027	203.8	Starr
21INR0359	Hickerson Solar	SOL	11/21/2025	311.1	Bosque
22INR0220	Lamkin Solar	SOL	08/08/2027	101.5	Comanche
22INR0239	Rockefeller Storage	BESS	06/01/2027	206.8	Schleicher
22INR0437	TORMES SOLAR	SOL	03/31/2027	382.12	Navarro
22INR0457	Anson BAT	BESS	08/01/2026	150.6	Jones
22INR0605	Camino Santiago Solar	SOL	02/18/2027	196.3	Milam
23INR0078	Shaw Solar	SOL	04/29/2026	124.65	Bandera
23INR0181	Starling Storage	BESS	05/15/2027	63.6	Gonzales
23INR0225	MRG GOODY SOLAR	SOL	05/02/2026	170.82	Lamar
23INR0479	Taormina Storage	BESS	05/26/2029	231.86	Bexar
23INR0538	Roadrunner Crossing BESS SLF	BESS	12/31/2025	150.4	Eastland
24INR0126	High Noon Storage	BESS	05/09/2028	94	Hill
24INR0181	Bynum Solar Project	SOL	12/01/2025	56	Coryell
24INR0188	Tehuacana Creek Solar SLF	SOL	03/10/2027	505.5	Navarro

Appendix C – Generation Added Cont.

GINR	Project Name	Fuel	Project COD	Capacity (MW)	County
24INR0189	Tehuacana Creek BESS SLF	BESS	03/10/2027	419	Navarro
24INR0305	MRG Goody Storage	BESS	05/02/2026	52.29	Lamar
24INR0355	Anatole Renewable Energy Storage	BESS	03/31/2027	207.82	Henderson
24INR0364	Pitts Dudik II	SOL	02/04/2026	30.23	Hill
24INR0386	Black & Gold Energy Storage	BESS	06/30/2027	254.58	Menard
24INR0453	Longfellow BESS I	BESS	01/31/2026	54.99	Pecos
24INR0455	Longfellow BESS II	BESS	01/31/2026	105.75	Pecos
24INR0493	Crowned Heron BESS 2	BESS	03/31/2026	154.2	Fort Bend
24INR0528	Blanquilla BESS	BESS	05/15/2026	200.78	Nueces
24INR0533	Padua Grid BESS Unit 2	BESS	03/15/2026	150.94	Bexar
24INR0584	Houston IV BESS	BESS	06/03/2026	164.6	Harris
25INR0018	Yellow Cat Wind	WIN	04/01/2027	262	Navarro
25INR0046	Blue Skies BESS	BESS	12/31/2027	306.26	Hill
25INR0199	Bonham Solar 1	SOL	08/31/2026	138.4	Limestone
25INR0229	OCI Cobb Creek Solar	SOL	12/01/2026	203.1	Hill

Appendix C – Generation Added Cont.

GINR	Project Name	Fuel	Project COD	Capacity (MW)	County
25INR0233	OCI Cobb Creek ESS	BESS	12/01/2026	201.6	Hill
25INR0391	Purple Sage BESS 1	BESS	05/30/2027	156	Collin
25INR0392	Purple Sage BESS 2	BESS	05/30/2027	156	Collin
26INR0034	Bracero Pecan Storage	BESS	04/01/2027	232	Reeves
26INR0296	Sherbino II BESS SLF	BESS	02/08/2026	77.37	Pecos
26INR0543	Three Canes Solar SLF	SOL	03/10/2027	333	Navarro
28INR0024	Padua Grid BESS Unit 3	BESS	05/15/2026	201.35	Bexar